Fakultas Ilmu Komputer

Universitas Indonesia

ASISTENSI DDAK WEEK 5

Sequential Circuit, Register, and Memory

Sequential Circuit

- Tabel Karakteristik Flip-Flop
- Analisis Sequential Circuit : State Table
- Analisis Sequential Circuit : State Diagram
- Excitation Table
- Mencari Flip-Flop Inputs berdasarkan State Diagram

RECALL : FLIP FLOPS

Karakteristik Table Flip Flop (jangan sampe lupa)

J	K	Q(t+1)	Comments
0	0	Q(t)	No change
0	1	0	Reset
1	0	1	Set
1	1	Q(t)'	Toggle

S	R	Q(t+1)	Comments
0	0	Q(t)	No change
0	1	0	Reset
1	0	1	Set
1	1	?	Unpredictable

D	Q(t+1)	
0	0	Reset
1	1	Set

T	Q(t+1)	
0	Q(t)	No change
1	Q(t)'	Toggle

SEQUENTIAL CIRCUITS ANALYSIS

- Dari sebuah diagram sirkuit, kita bisa menganalisa input dan outputnya menggunakan **State Table dan State Diagram**
- Untuk mendapatkan analisis tersebut, kita bisa menggunakan **State Equations dan Output Functions**
- Notasi dari **present state dan next state** dari suatu flip flop bisa menggunakan **A(t) dan A(t+1) atau A dan A+**
- **Di asistensi ini, kita pakai notasi A dan A+**

SEQUENTIAL CIRCUITS ANALYSIS

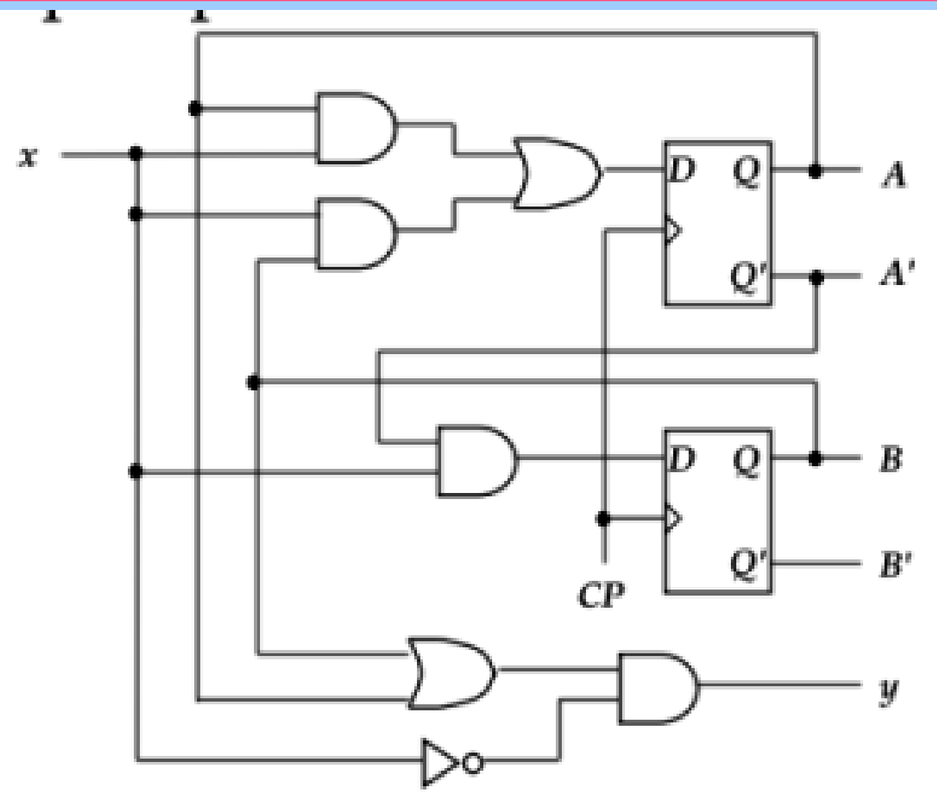


Figure 1

State equations:

DA = A+ = A.x + B.x

B+ = A'.x

Output function:

y = (A+B).x'

SEQUENTIAL CIRCUITS ANALYSIS

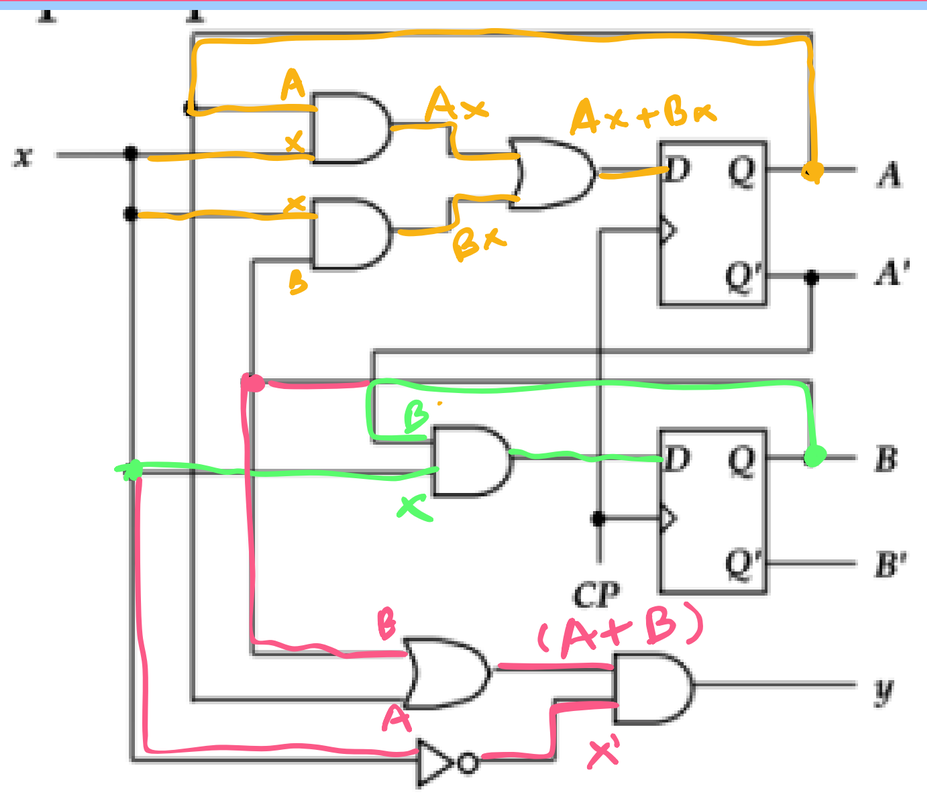


Figure 1

State equations:

DA = A+ = A.x + B.x

B+ = A′.x

Output function:

y = (A+B).x′

STATE TABLE

- Mirip dengan **truth table**
- Input dan present state ada di sebelah kiri
- Output dan next state ada di sebelah kanan
- m = jumlah flip flop, n = jumlah input. Maka jumlah baris pada state table adalah 2^{m+n}

CONTOH STATE TABLE

Present State		Input	Next State		Output
A	B	x	A^+	B^+	y
0	0	0			
0	0	1			
0	1	0			
0	1	1	1	1	0
1	0	0			
1	0	1			
1	1	0			
1	1	1			

State equations:

DA = A+ = A.x + B.x

B+ = A'.x

Output function:

y = (A+B).x'

Contoh: A = 0, B = 1, x = 1

- A+ = 0 . 1 + 1 . 1 = 0 + 1 = 1
- B+ = (0)' . 1 = 1 . 1 = 1
- y = (0 + 1) . (1)' = 1 . 0 = 0

CONTOH STATE TABLE

Present State		Input	Next State		Output
A	B	x	A^+	B^+	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0

State equations:

DA = A+ = A.x + B.x

B+ = A'.x

Output function:

y = (A+B).x'

Contoh: A = 0, B = 1, x = 1

- A+ = 0 . 1 + 1 . 1 = 0 + 1 = 1
- B+ = (0)' . 1 = 1 . 1 = 1
- y = (0 + 1) . (1)' = 1 . 0 = 0

JENIS STATE TABLE

Full Table

Present State		Input	Next State		Output
A	B	x	A^+	B^+	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0

Compact Table

Present State	Next State		Output	
	$x=0$	$x=1$	$x=0$	$x=1$
AB	A^+B^+	A^+B^+	y	y
00	00	01	0	0
01	00	11	1	0
10	00	10	1	0
11	00	10	1	0

STATE DIAGRAM

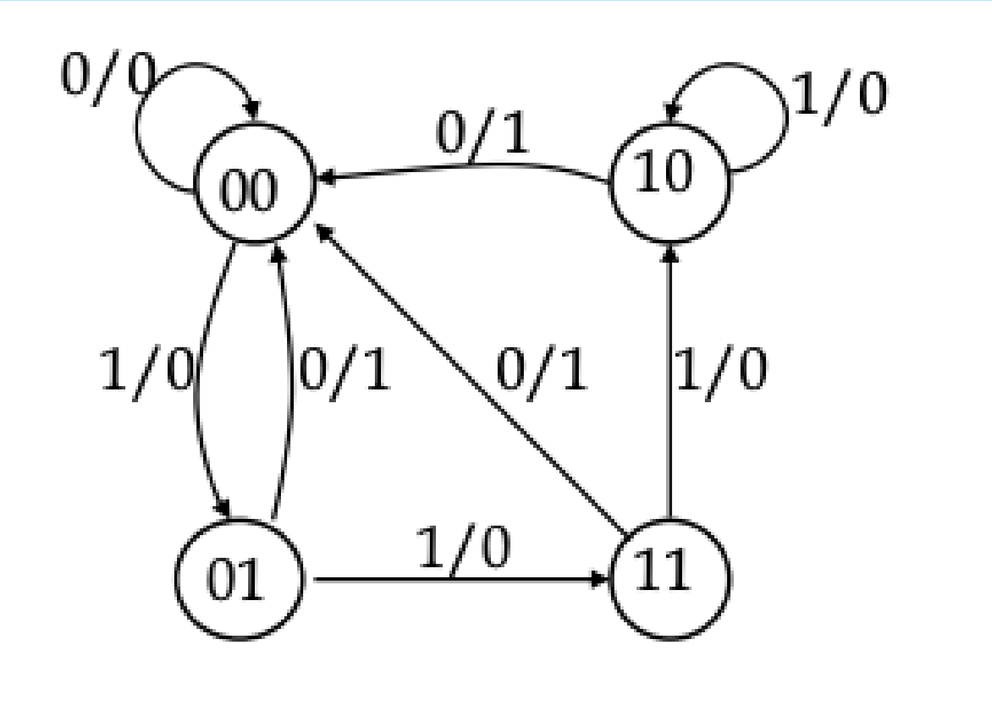
Dari state table tadi,
kita bisa bikin diagram
baru bernama **State
Diagram**

CONTOH STATE DIAGRAM

Present State	Next State		Output	
	$x=0$	$x=1$	$x=0$	$x=1$
AB	A^+B^+	A^+B^+	y	y
00	00	01	0	0
01	00	11	1	0
10	00	10	1	0
11	00	10	1	0

- Setiap state digambarkan menggunakan lingkaran
- Anak panah mengarahkan ke next state
- Angka yang ada di anak panah adalah x/y

State Diagram



EXCITATION TABLE

Selain characteristic table flip-flop, ada juga excitation table flip flop buat **nentuin present state dan next state berdasarkan input dari masing-masing Flip Flop**

Q	Q^+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

JK Flip-flop

Q	Q^+	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

SR Flip-flop

Q	Q^+	D
0	0	0
0	1	1
1	0	0
1	1	1

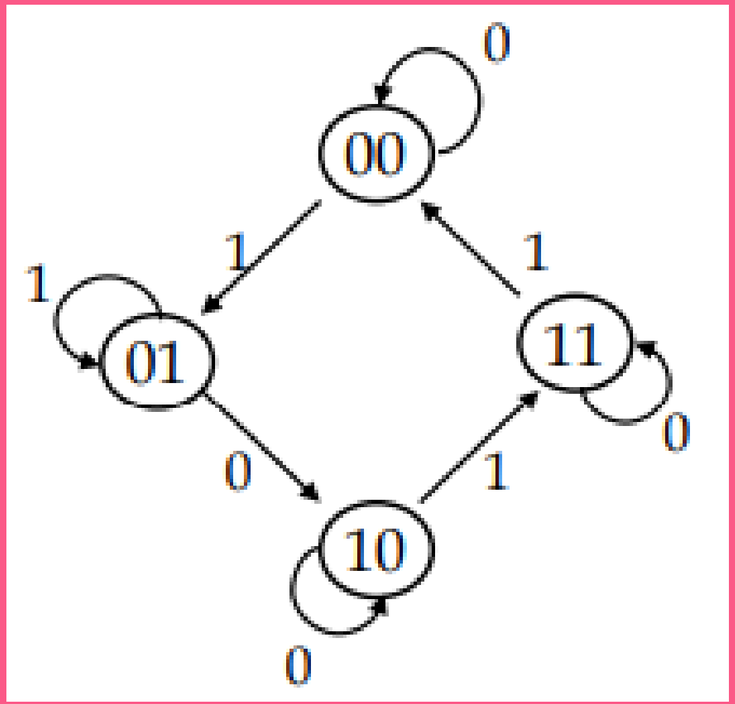
D Flip-flop

Q	Q^+	T
0	0	0
0	1	1
1	0	1
1	1	0

T Flip-flop

MENCARI FLIP-FLOP INPUTS BERDASARKAN STATE DIAGRAM

Dari state diagram berikut, coba **buat sequential circuit menggunakan JK Flip Flop!**

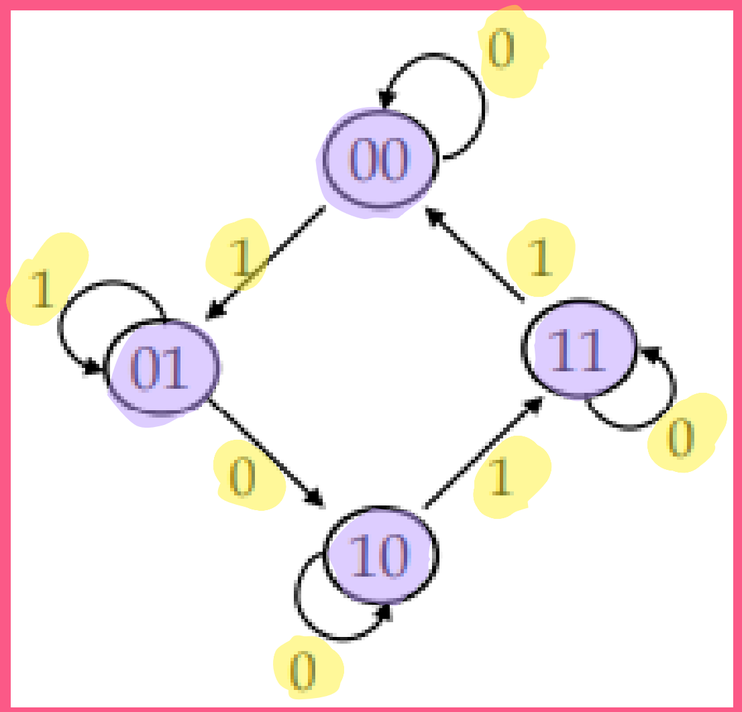


Steps:

- **Buat compact state table**
- **Buat full state table** (Gunakan JK Excitation Table sebagai rujukan untuk membuat state table)
- **Cari State Equation dari K-Map**
- **Bangun sequential circuit berdasarkan State Equation**

MENCARI FLIP-FLOP INPUTS BERDASARKAN STATE DIAGRAM

Step 1: Buat state table

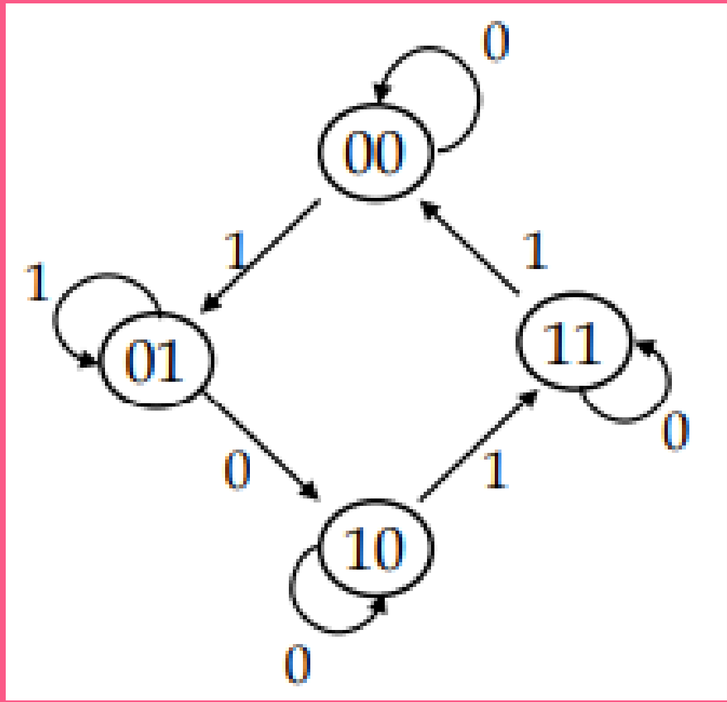


Present State	Next State	
	$x=0$	$x=1$
AB	A^+B^+	A^+B^+

Compact State Table

MENCARI FLIP-FLOP INPUTS BERDASARKAN STATE DIAGRAM

Step 1: Buat state table



Present State	Next State $x=0$	Next State $x=1$
AB	A^+B^+	A^+B^+
00	00	01
01	10	01
10	10	11
11	11	00

Compact State Table

MENCARI FLIP-FLOP INPUTS BERDASARKAN STATE DIAGRAM

Step 2: Buat full state table

Present State	Next State $x=0$	Next State $x=1$
AB	A^+B^+	A^+B^+
00	00	01
01	10	01
10	10	11
11	11	00

Q	Q^+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

JK Flip-flop's excitation table.

Present state		Input	Next state		Flip-flop inputs			
A	B	x	A^+	B^+	JA	KA	JB	KB
0	0	0	0	0				
0	0	1	0	1				
0	1	0	1	0				
0	1	1	0	1	0	X	X	0
1	0	0	1	0				
1	0	1	1	1				
1	1	0	1	1				
1	1	1	0	0				

Full State Table

MENCARI FLIP-FLOP INPUTS BERDASARKAN STATE DIAGRAM

Step 2: Buat full state table

Present State	Next State $x=0$	Next State $x=1$
AB	A^+B^+	A^+B^+
00	00	01
01	10	01
10	10	11
11	11	00

Q	Q^+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

JK Flip-flop's excitation table.

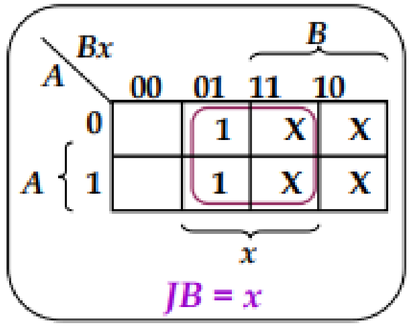
Present state		Input	Next state		Flip-flop inputs			
A	B	x	A^+	B^+	JA	KA	JB	KB
0	0	0	0	0	0	X	0	X
0	0	1	0	1	0	X	1	X
0	1	0	1	0	1	X	X	1
0	1	1	0	1	0	X	X	0
1	0	0	1	0	X	0	0	X
1	0	1	1	1	X	0	1	X
1	1	0	1	1	X	0	X	0
1	1	1	0	0	X	1	X	1

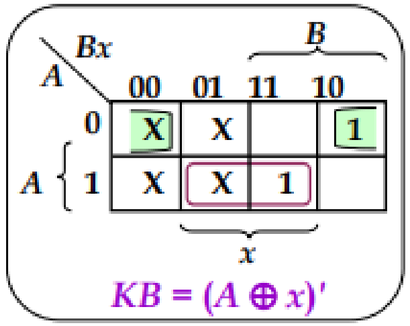
Full State Table

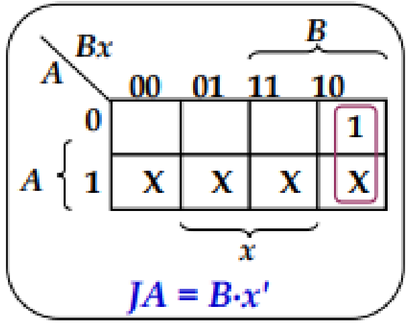
MENCARI FLIP-FLOP INPUTS BERDASARKAN STATE DIAGRAM

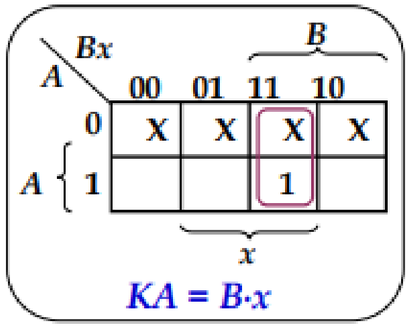
Step 3: Cari State Equation dari K-Map

Present state		Input	Next state		Flip-flop inputs			
A	B	x	A^+	B^+	JA	KA	JB	KB
0	0	0	0	0	0	X	0	X
0	0	1	0	1	0	X	1	X
0	1	0	1	0	1	X	X	1
0	1	1	0	1	0	X	X	0
1	0	0	1	0	X	0	0	X
1	0	1	1	1	X	0	1	X
1	1	0	1	1	X	0	X	0
1	1	1	0	0	X	1	X	1







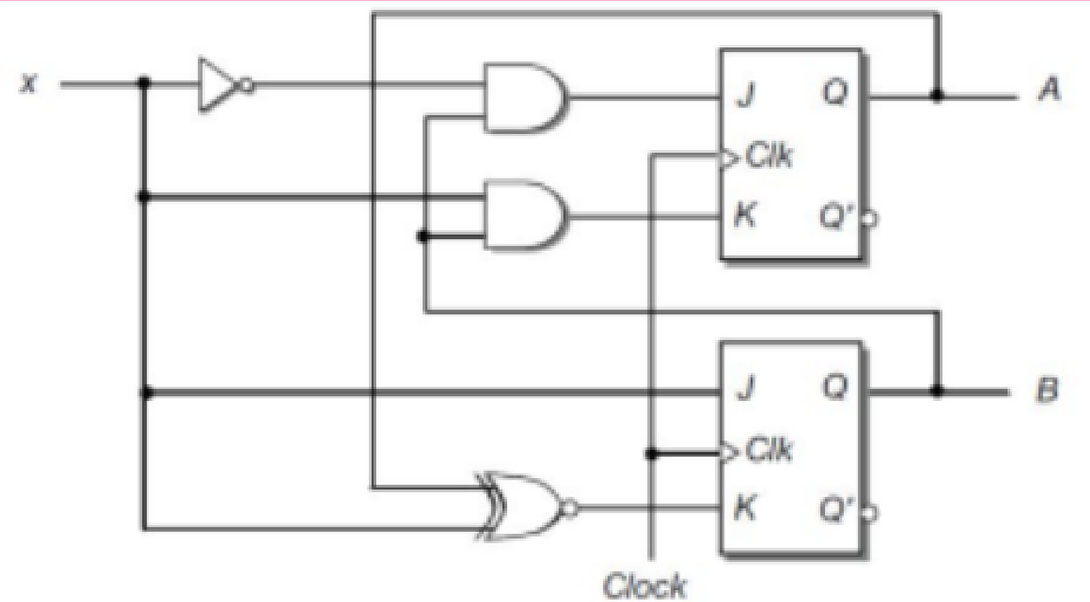


MENCARI FLIP-FLOP INPUTS BERDASARKAN STATE DIAGRAM

Step 4: Bangun sequential circuit berdasarkan State Equation

State Equation

$JA = B \cdot x'$	$JB = x$
$KA = B \cdot x$	$KB = (A \oplus x)'$



Sequential circuit diagram

REGISTERS

INTRODUCTION

- Flip-Flop adalah rangkaian digital yang dapat menyimpan 1 bit data (0 atau 1).
- Flip-Flop digunakan sebagai elemen dasar penyimpanan data dalam sistem digital.

- Setiap Flip-Flop menyimpan 1 bit = 1 bit memory cell
- **Kumpulan beberapa Flip-Flop yang bekerja bersama disebut Register**

n-bit Register

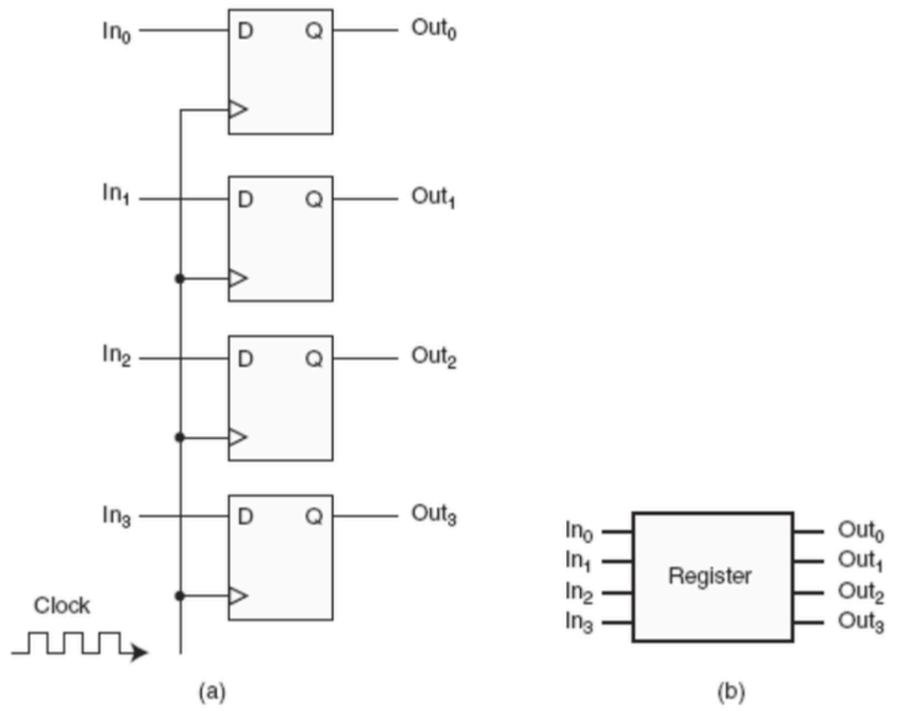


FIGURE 3.23 a) A 4-Bit Register
b) A Block Diagram for a 4-Bit Register

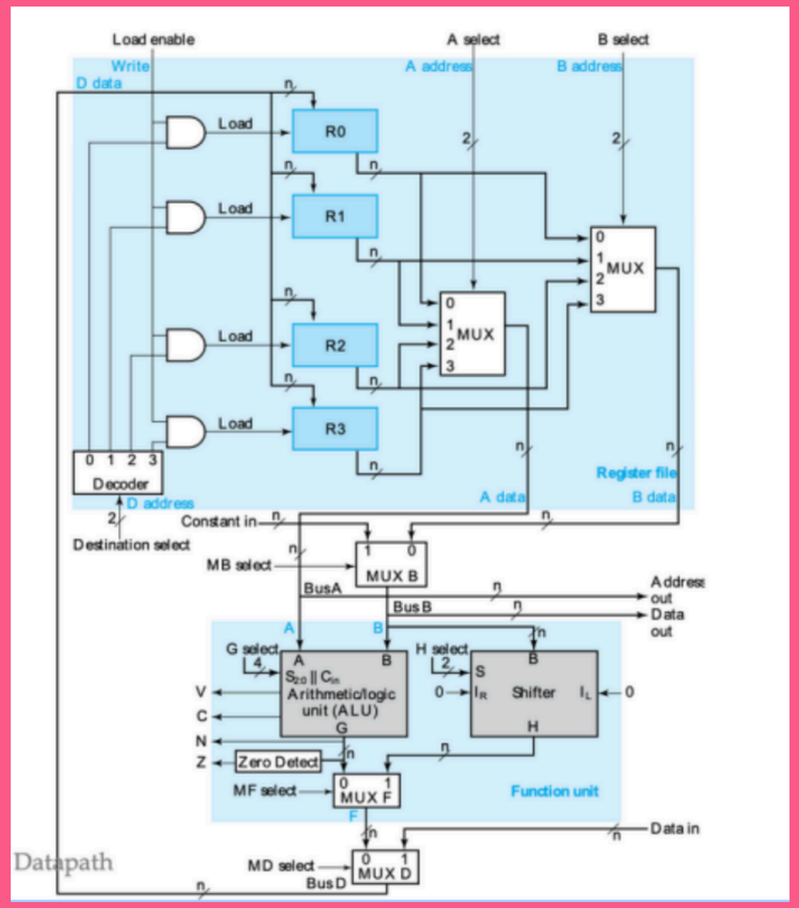
Register digunakan untuk:

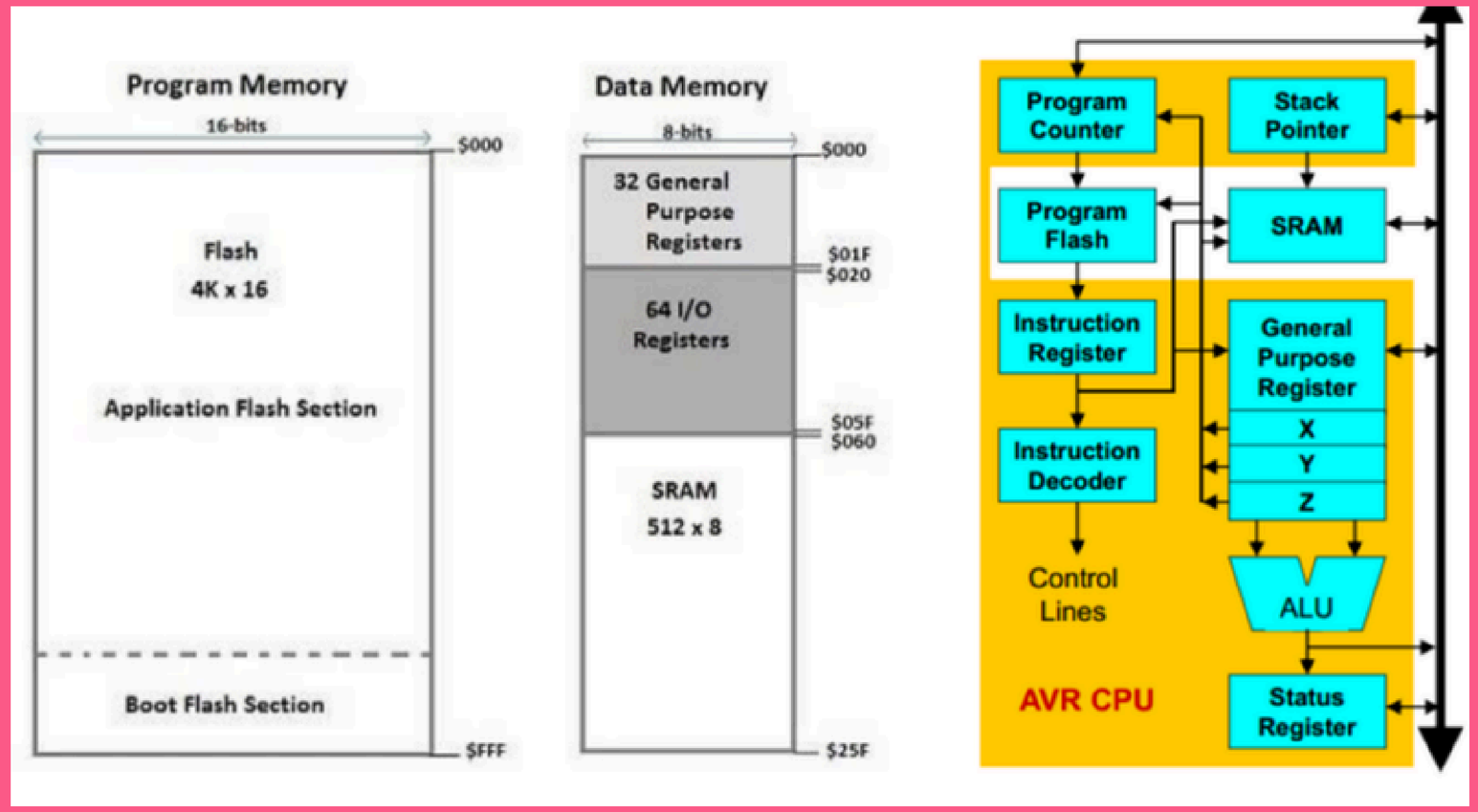
- **Menyimpan data sementara**
- **Menyimpan hasil operasi**
- **Menyimpan instruksi dalam CPU**

Jika terdapat n Flip-Flop, maka disebut n-bit register.

Contoh:
4 Flip-Flop → 4-bit register
8 Flip-Flop → 8-bit register
16 Flip-Flop → 16-bit register

n-bit Register Example





COUNTER REGISTER

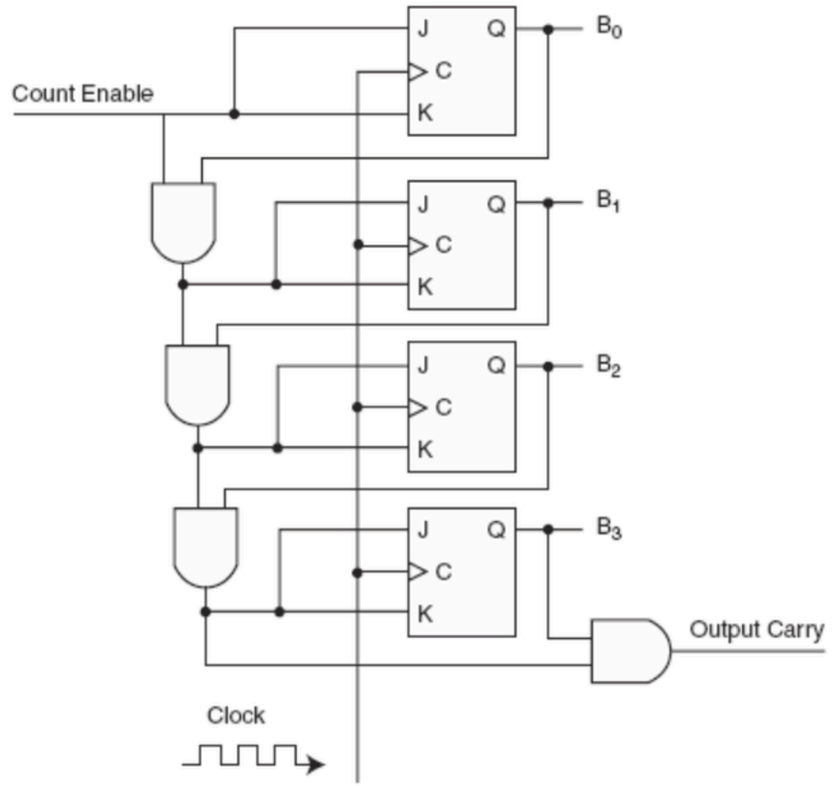


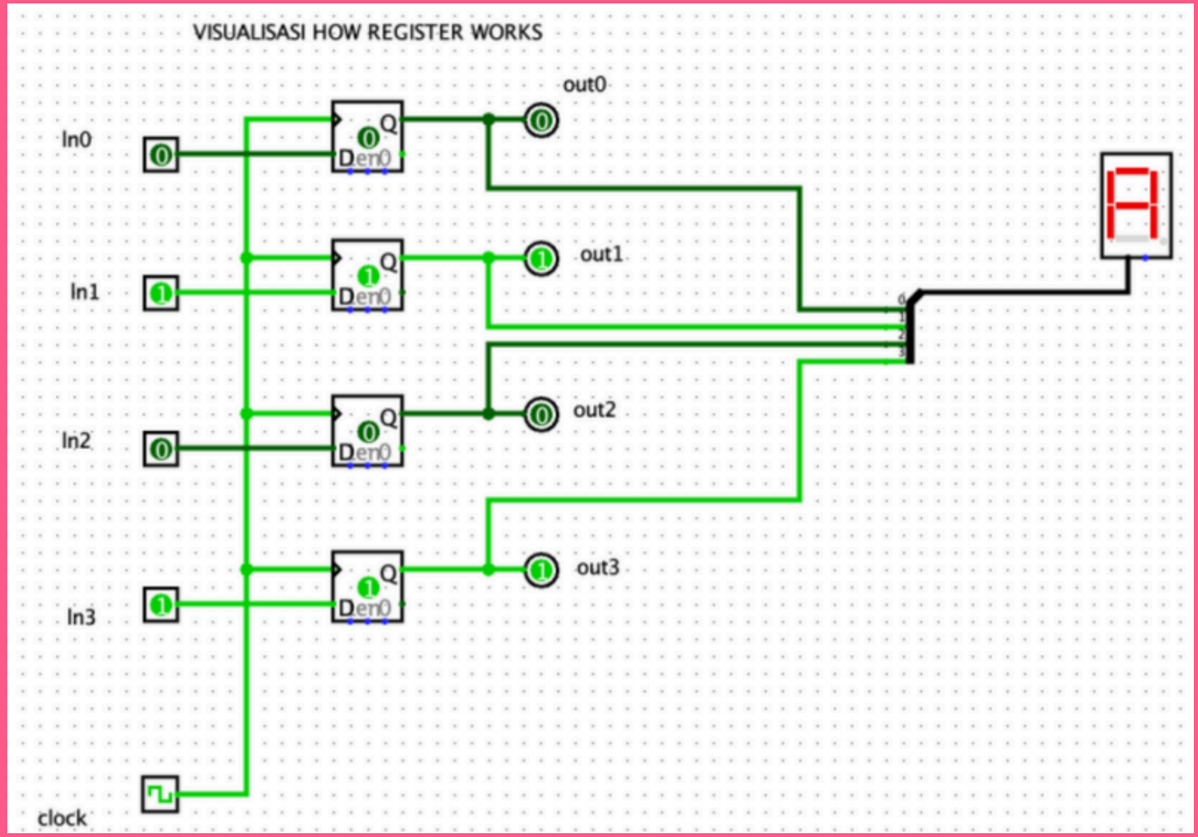
FIGURE 3.24 A 4-Bit Synchronous Counter Using JK Flip-Flops

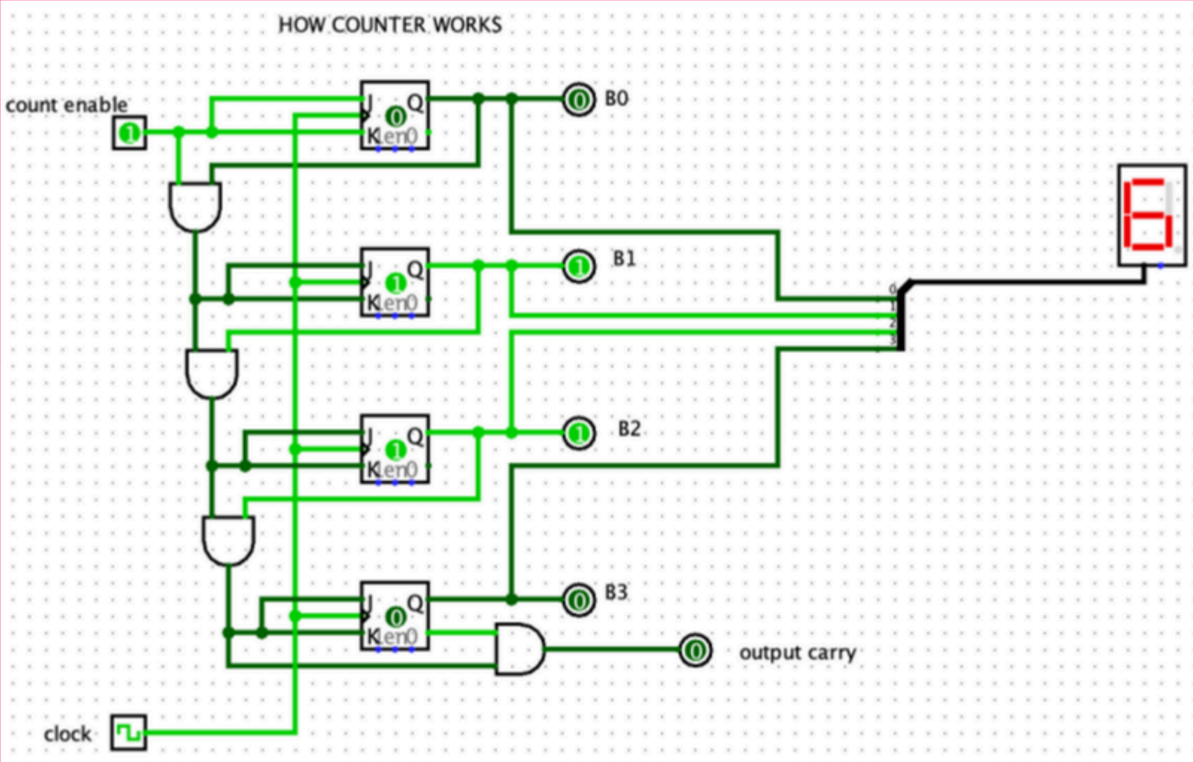
Counter adalah register:

- Nilai pada counter akan berubah setiap clock secara otomatis mengikuti urutan tertentu (cycle)
- Dibangun menggunakan Flip-Flop

=> counter (literally)

REGISTER PADA LOGISM





TYPICAL QUESTION

Sebuah register 16-bit menggunakan berapa flip-flop?
ans: 16 flip-flop

TYPICAL QUESTION

Manakah pernyataan dibawah ini yang BENAR terkait Register
Note: jawaban dapat lebih dari satu

A. 16-Bit Register membutuhkan 16 buah Flip-flop

B. 16-Bit Register membutuhkan 4 buah Flip-flop

C. Output register hanya tergantung dari present input

D. Output register hanya tergantung dari present input dan ouput/state sebelumnya → register itu sequential

E. Input register bersifat Synchronous → karena flip-flop

F. Input register bersifat Asynchronous

MEMORY

INTRODUCTION

1 byte = 8 bits
1 kilobyte (KB) = 2^{10} bytes
1 megabyte (MB) = 2^{20} bytes
1 gigabyte (GB) = 2^{30} bytes
1 terabyte (TB) = 2^{40} bytes

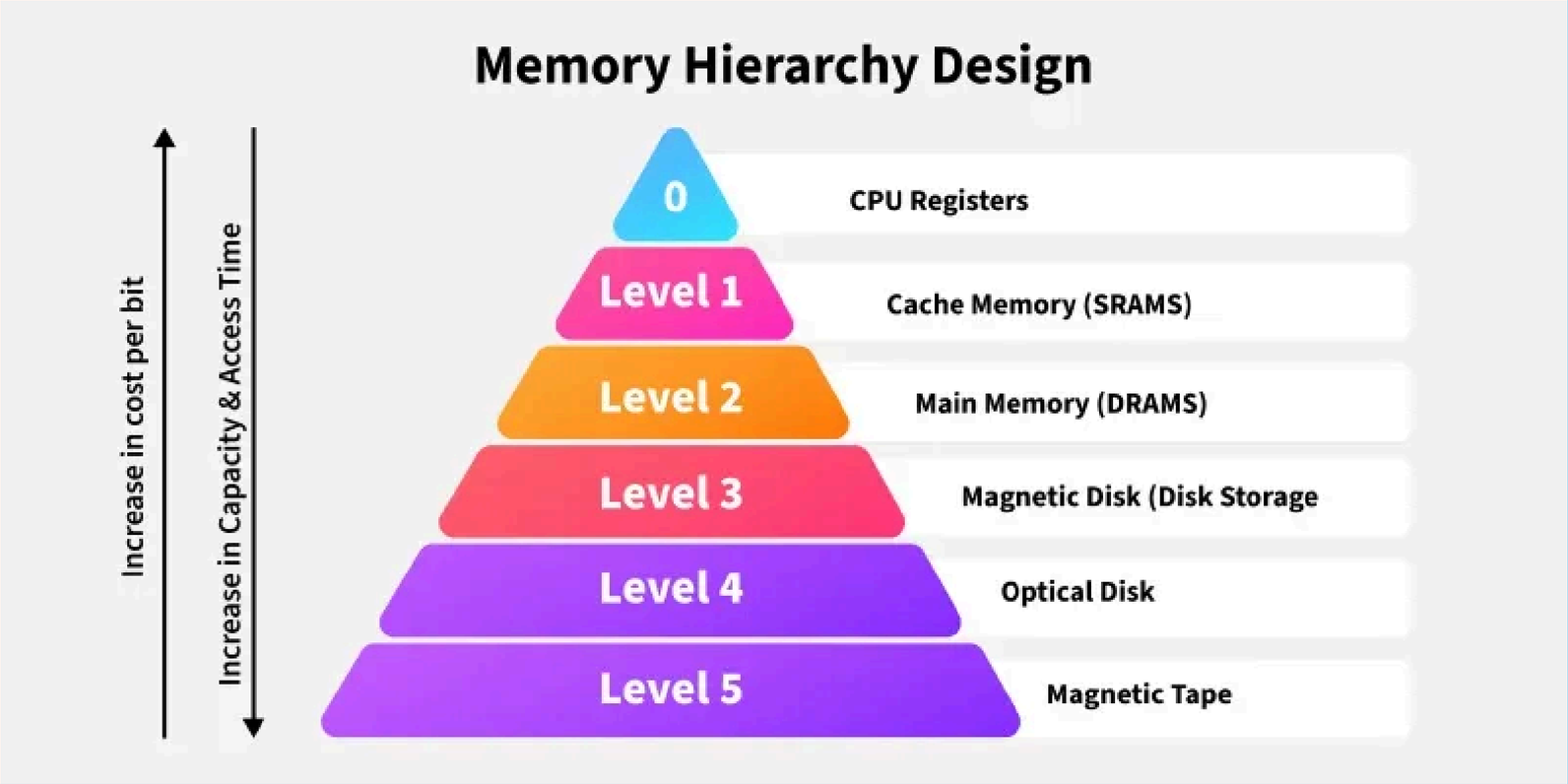
INTRODUCTION

Memori berfungsi untuk menyimpan program dan data.

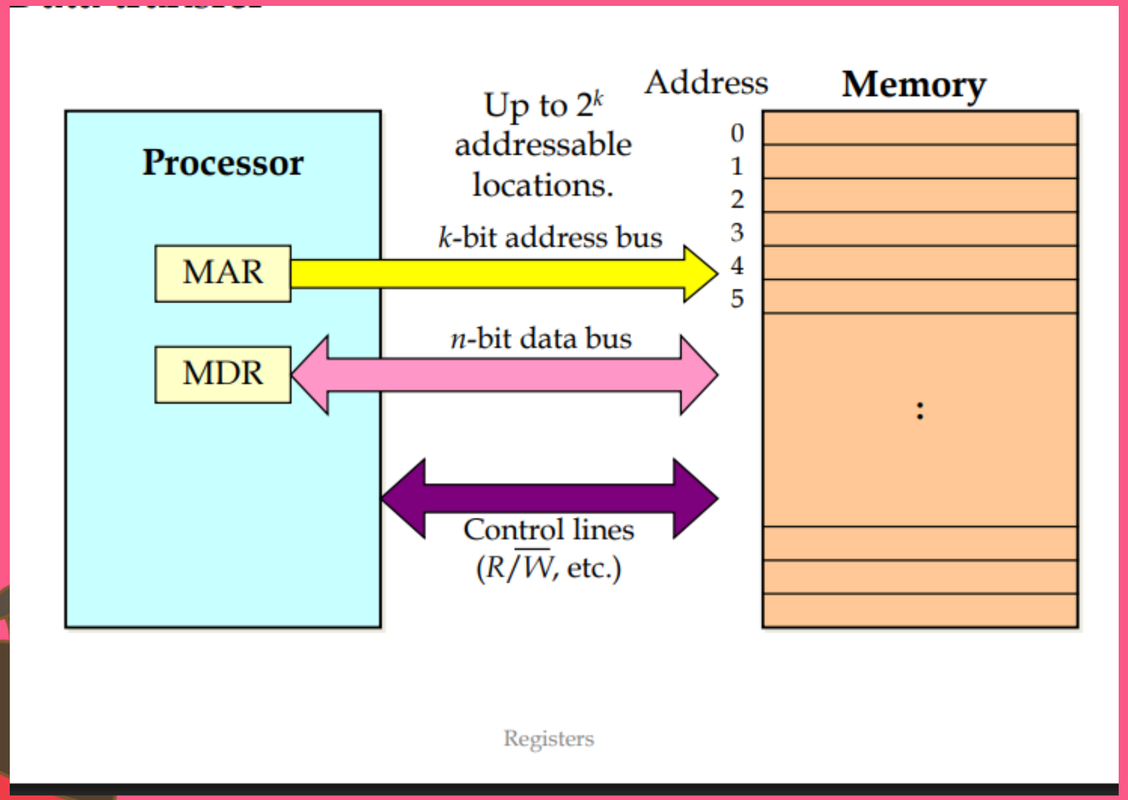
Memori yang ideal memiliki akses cepat, non-volatile, kapasitas besar, dan murah

Tetapi tidak ada memori yang ideal

HIERARCHY


Memory Hierarchy Design
Increase in cost per bit
Increase in Capacity & Access Time
0
CPU Registers
Level 1
Cache Memory (SRAMS)
Level 2
Main Memory (DRAMS)
Level 3
Magnetic Disk (Disk Storage
Level 4
Optical Disk
Level 5
Magnetic Tape

DATA TRANSFER



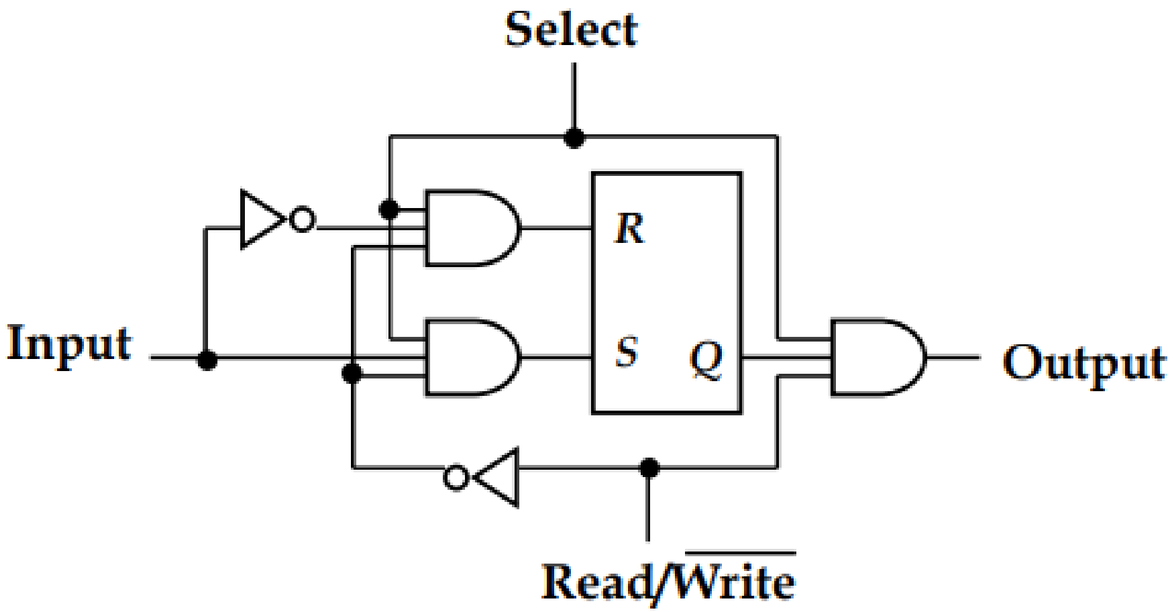
Memory menyimpan informasi dalam bentuk words.

Suatu memory unit memiliki 2^k alamat (ditunjuk oleh k-bit address bus) dan memiliki word sepanjang n-bits.

Control lines menentukan operasi yang akan dilakukan (read atau write)

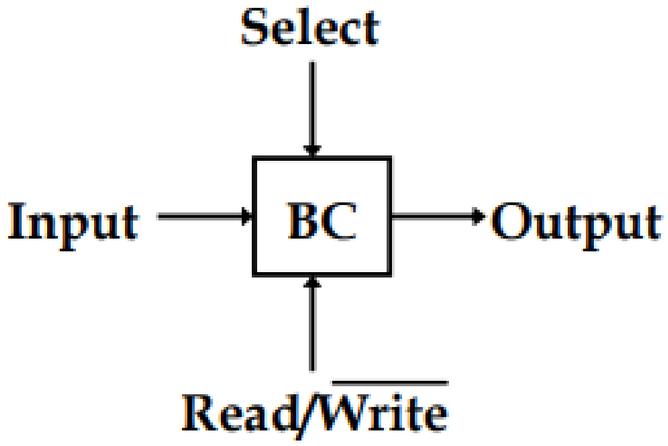
Memory Enable	*Read/Write*	Memory Operation
0	X	
1	0	WRITE
1	1	READ

RAM TYPES



Logic diagram

STATIC



Block diagram

DYNAMIC

LATIHAN SOAL

- $2M \times 32$ memory module
  - Using $512K \times 8$ memory chips.

Cara menentukan formasi array:

Berapa kolom → 32/8 = 4

Berapa baris → 2M/512K = $2^{21}/2^{19} = 2^2 = 4$

Total chip = 4 * 4 = 16 chip

Jumlah address dalam satu chip memori =512K=2^{19}

Setiap entry menyimpan data sebesar: 8 bit

Total bit encoder → 2M/512K = $2^{21}/2^{19} = 2^2 = 2$ bit

ANY QUESTIONS?

LINK RECORDING ASISTENSI

DISINI

Kode sandi: ?VE2&F70

THANK YOU AND GOOD LUCK

CHEL, RMH, RAF